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Generative artificial intelligence for enzyme design: Recent advances in models and applications

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Enzyme catalysis is a key enabling technology for green and sustainable production of chemicals. Developing suitable enzymes is at the heart of this technology, which is currently changing by Artificial Intelligence (AI) such as machine learning. AI-based methods were used for enzyme discovery and design. We review the recent advances in generative AI models for enzyme design, with a particular focus on those that have been validated by experiments. Furthermore, we discuss the applications of the enzymes designed by generative AI, including artificial luciferases, non-heme iron (II)-dependent oxygenases, and P450 enzymes. We provide our opinions on several current issues encountered in computational enzyme design. With the fast development of new generative models in enzymes and the implementation of these models by the research community, we believe that the precise design of efficient enzymes with new catalytic functions and/or potential industrial applications will be a mature method in the near future.

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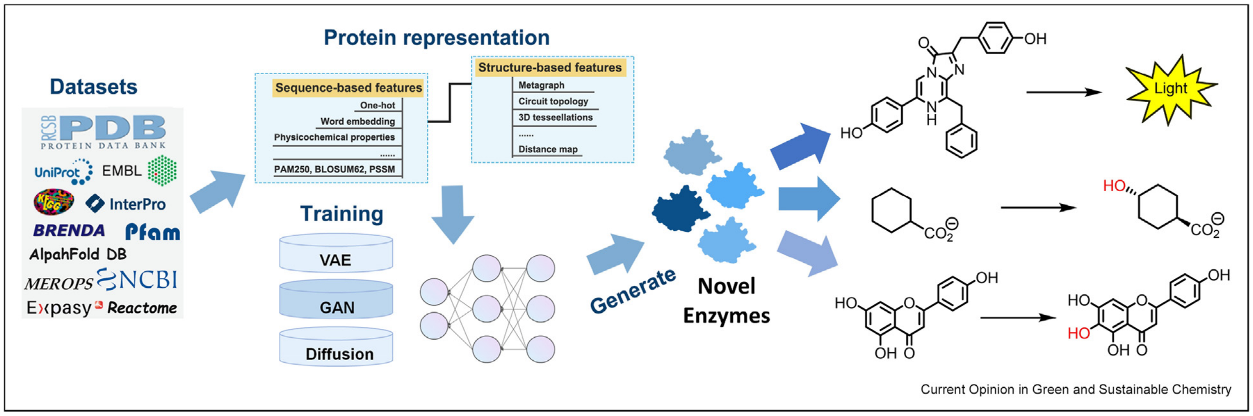
Introduction

Enzymes are the “core chips” of biological production or conversion of chemicals in a green and sustainable manner via single- or multi-step enzyme catalysis [1–6], as well as cell-based metabolic engineering and synthetic biology [7–10]. To transfer natural enzymes into industrial biocatalysts, a widely used key enabling technology is directed evolution, a mainly experimental approach, by applying mutation-screening cycles on the protein level (*i.e.*, molecular breeding) without a deep understanding of the molecular mechanism [11–13]. Although many successes for the engineering of enzymes have been reported by academia and industry, directed evolution is still time-, labor-, equipment- and resource-consuming, often limited to natural enzymes as starting templates, frequently trapped in some local optimum of evolution path, and it is challenging to create new functions. In the last decade, due to the advances in catalytic mechanisms and physics-based simulations, semi-rational or even rational approaches have been developed and applied for engineering or redesign of improved enzyme variants [14–17], yet this is still largely limited to natural reactions and natural sequences as the prototypes.

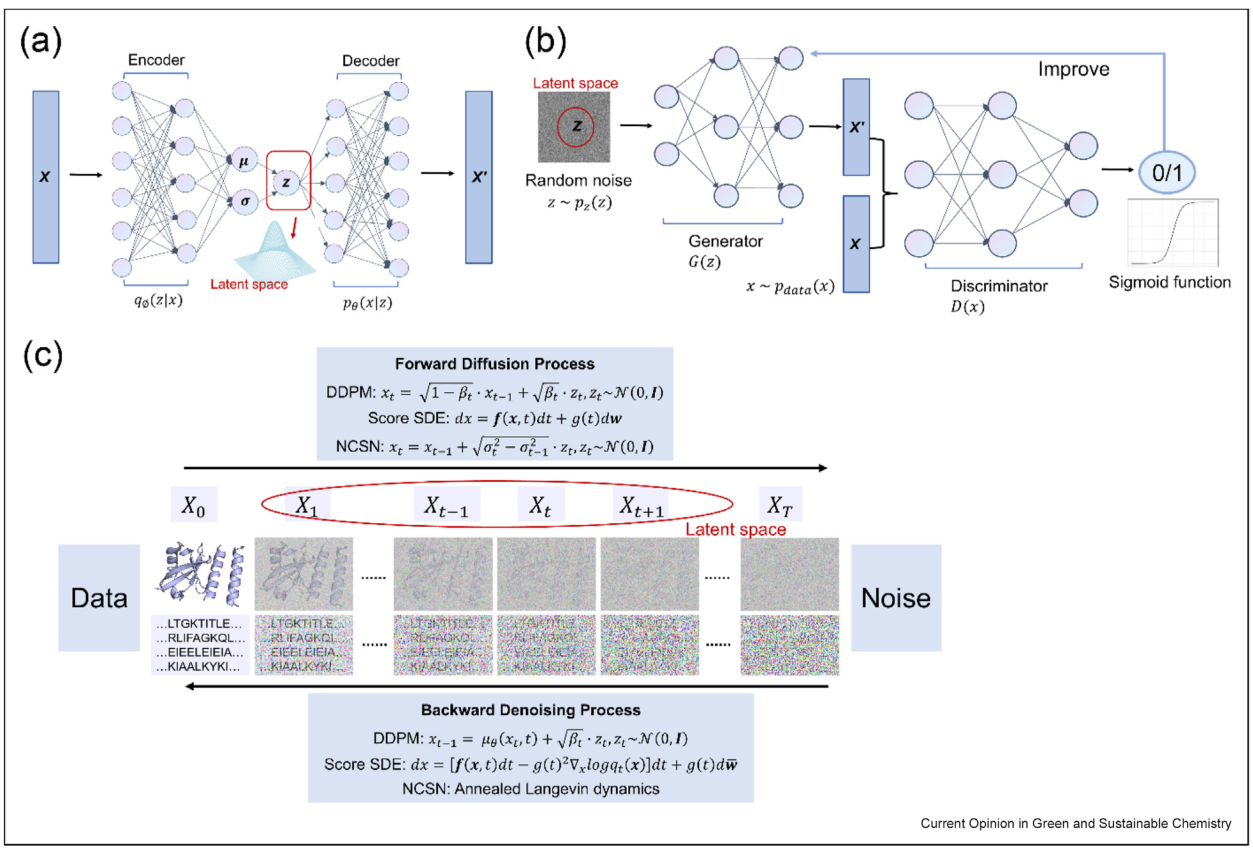
Recently, breakthroughs in artificial intelligence together with rapid accumulation of big data of protein sequences and structures provide a unique data-driven approach for enzyme discovery, engineering, and design [18–22]. This is obvious from the Nobel prize in Chemistry awarded in 2024 to David Baker for his achievements in computational protein design and to Demis Hassabis and John Jumper for protein structure prediction using AlphaFold [23]. These tools are very important for machine-learning assisted data-driven enzyme engineering, which has been developed to improve the stability, activity, and selectivity of many industrially used enzymes (*e.g.*, PET hydrolases [24,25], imine reductases [26], and amine transaminases [27]) which had been intensively reviewed very recently [28–30]. Besides the prediction of mutations, AI-based methods could also generate new sequences for protein design [31–33]. Herein, we focus our discussions on recent developments of AI-based generative models for the creation of novel enzymes (instead of simple variants of existing enzymes) for production of chemicals (Fig. 1).

Figure 1



Development of generative AI models for creating novel enzymes for various applications.

Figure 2



Overview of the three generative model architectures covered in this review. (a) Variational autoencoders (VAEs) comprise an encoder that parameterizes the distribution over the latent variables *z* and a decoder that reconstructs inputs from samples drawn from this distribution. (b) Generative adversarial networks (GANs) consist of a generator that creates realistic examples to deceive the discriminator, and a discriminator that differentiates between real and generated examples. (c) Diffusion model consists of two processes: the forward process transforms data into noise, while the reverse process learns to transform the noise back into data. We highlight three alternative formulations of diffusion models: stochastic differential equations (Score SDEs), denoising diffusion probabilistic models (DDPMs) and noise conditioned score networks (NCSNs).

Generative models for enzyme design

Owing to the advances in hardware and deep neural networks, Deep Generative Models (DGMs) such as Variational Autoencoder (VAE), Generative Adversarial Network (GAN), and diffusion model (Fig. 2) have achieved significant breakthroughs in image generation, chatbot, and music composition [34]. DGMs can efficiently explore latent sequence spaces and generate sequences with desired functionalities when applied to functional enzymes. Here, we highlight the improved algorithms and exceptional performance of three representative DGMs in enzyme design. Details on each experimentally validated model used in enzyme design, along with their specific information, can be found in Table 1.

Variational autoencoder models

Evolving from traditional autoencoders, VAEs can be understood as two interconnected, yet independently parameterized models: the encoder, or recognition model, and the decoder, or generative model. When processing protein sequences, the encoder compresses the original protein sequence to low-dimensional latent vectors, while the decoder reconstructs the original input sequences from these latent vectors (Fig. 2a). Specifically, the vector of input sequences $\boldsymbol{x}$ can be mapped to a latent representation $\boldsymbol{z}$ using the probabilistic encoder $q_{\varnothing}(\boldsymbol{z}|\boldsymbol{x})$, which is parameterized by $\varnothing$ and assumed to follow a Gaussian distribution. Then, $\boldsymbol{z}$ can be used to reconstruct the sequence, with the decoder $p_{\theta}(\boldsymbol{x}|\boldsymbol{z})$ parameterized by θ. VAE is trained to minimize the distance between the pristine data and the reconstructed data by maximizing the *evidence lower bound* (ELBO), $\mathscr{L}(\varnothing;\boldsymbol{x})$, as follows:

$$ELBO = E_{q_{\varnothing}(\boldsymbol{z}|\boldsymbol{x})}[log\, p_{\theta}(\boldsymbol{x}|\boldsymbol{z})] - D_{KL}[q_{\varnothing}(\boldsymbol{z}|\boldsymbol{x})\|p(\boldsymbol{z})]$$

here, $\varnothing$ and θ represent the parameterized distribution of VAE probabilistic encoder-decoder structure, the term $E_{q_{\varnothing}(\boldsymbol{z}|\boldsymbol{x})}[log\, p_{\theta}(\boldsymbol{x}|\boldsymbol{z})]$ represents the expected conditional log-likelihood, $p(\boldsymbol{z})$ denotes the prior Gaussian distribution, and $D_{KL}[q_{\varnothing}(\boldsymbol{z}|\boldsymbol{x})\|p(\boldsymbol{z})]$ represents Kullback-Leibler (KL) divergence between real and the approximate posterior [35,36].

In terms of enzyme design, Hawkins-Hooker et al. introduced two VAE models — MSA VAE, which uses aligned sequence input, and AR-VAE, which uses raw sequence input — to generate novel functional variants of the *luxA* bacterial luciferase and experimentally validated their luminescent activity (Fig. 3a) [37]. Besides, ProT-VAE [36], which combines the advantages of variational autoencoders with the feature representations of transformers, was tested on a *de novo* sequence design task for phenylalanine hydroxylase (PAH), resulting in the design and experimental validation of new PAH sequences with over 100 mutations and up to a 2.5 × increase in activity compared to human PAH (Fig. 3b). Kohout et al. introduced evolutionary trajectories generated by an optimized deep-learning framework of VAE, which were used to generate ancestral sequences of haloalkane dehalogenase, demonstrating wild-type stability and activity in soluble variants (Fig. 3c) [38]. When incorporating functional measurements within a semi-supervised learning framework, the ProtWave-VAE model [39] — which combines an information-maximizing VAE with a dilated convolutional encoder and a WaveNet autoregressive decoder — can reshape the latent space to induce a gradient in the property of interest and maintain its generative capabilities when trained on chorismate mutase protein family (Fig. 3d). Furthermore, ABACUS-R [40] — an encoder-decoder network trained using a multitask learning strategy based on the Transformer architecture — consists of two components: an encoder-decoder pretrained to infer the side-chain type of a central residue from its local 3D environment, and an iterative process that refines side-chain assignments starting from random sequences. This model can accurately predict the side-chain type of central residues, with most designed sequences being successfully expressed in *Escherichia coli*, purified, and experimentally verified to exhibit stable folded structures.

Generative adversarial network models

GANs are structured probabilistic models consisting of a generator network (G) and a discriminator network (D), where the generator aims to deceive the discriminator by learning the distribution of real data to generate highly realistic data from a latent vector $\boldsymbol{z}$, while the discriminator functions as a binary classifier, tasked with distinguishing generated data from real data as accurately as possible (Fig. 2b). While training GAN models, the generator is initially fixed to train the discriminator for improving its distinguishing accuracy, and then the discriminator is fixed to train the generator to enhance its data generation ability. Throughout this process, which resembles a minimax zero-sum game, both networks optimize themselves to achieve their respective objectives, as each has its own objective function. Specifically, the discriminator (D) aims to maximize its cost function, while the generator (G) seeks to minimize its cost function, which is expressed as follows:

$$V(D,G) = E_{x\sim p_{data}(x)}[logD(x)] + E_{z\sim p(z)}[\log(1-D(G(z)))]$$

where $logD(x)$ represents the loss function for the discriminator network and $\log(1-D(G(z)))$ is the loss function for the generator network.

In the realm of protein studies, a variant of GAN model named ProteinGAN [41] directly learns the diversity of natural protein sequences and generates functional new sequences, expanding the protein sequence space for catalytic design. The model was tested using malate

Table 1

Experimentally validated generative models developed for protein (enzyme) design in recent years.

Model name	Input	Model type	Release date	Function in protein design	Training set	Experimentally validated enzymes	Code availability	Ref
MSA VAE and AR-VAE	Protein sequence	VAE	2021	Generate novel, functional variants	~70,000 luciferase-like oxidoreductases acquired from UniProt	*luxA* bacterial luciferase	https://github.com/alex-hh/deepprotein-generation	[37]
ProT-VAE	Protein sequence	VAE	2023	Generate protein sequence	Pretrained using ProtT5	Phenylalanine hydroxylase	No	[36]
ProtWave-VAE	Protein sequence	VAE	2023	Design novel protein sequences and generate synthetic sequences	1759 DHFR, 4974 GTPase, 814 β-lactamase, 1344 S1A proteases	Src homology 3	No	[39]
RecGen	Protein sequence	VAE	2022	Sequence prediction	1.1 million cre-like recombinases with PacBio HiFi sequencing	Recombinases	https://github.com/ltschmitt/RecGen	[68]
ABACUS-R	Protein structure	Encoder−decoder model	2022	Protein sequence generation	A set of non-redundant protein data bank (PDB) structures	PDB IDs: 1r26, 1cy5 and 1ubq	https://doi.org/10.24433/CO.3351944.v1	[40]
ProteinGAN	Protein sequence	GAN	2021	Creates new, highly diverse sequence variants	16,898 bacterial MDH sequences downloaded from UniProt	Malate dehydrogenase	https://github.com/Biomatter-Designs/ProteinGAN	[41]
HelixGAN	Protein sequence	GAN	2023	Generate α-helix structures from scratch	3 million helical structures captured from PDB	MD simulations validated the stability of the designed D-GLP1_1 analog	https://github.com/xxiexuezhi/helix_gan	[42]
ProteinSGM	Protein structure	Diffusion model	2022	Protein scaffold generation	CATH 4.3 non-redundant S40 dataset	PDB 2KL8 and 7MRX are reference structures of test cases. Mdm2 scaffold is designed	https://gitlab.com/mjslee0921/proteinsgm	[56]
Chroma	Protein structure	Diffusion model	2023	Proteins and protein complexes generation	PDB queried on 20 March 2022, UniProt 2022_01, PFAM 35	Generated 310 proteins for expression and structural characterization	https://github.com/generatebio/chroma	[48]
RFdiffusion	Protein structure	Diffusion model	2023	Protein backbones generation	Protein structures obtained from PDB	Retroaldolase, oxidoreductase, transferase, lyase, hydrolase, isomerase	https://github.com/RosettaCommons/RFdiffusion	[49]
RFdiffusionAA	Protein structure	Diffusion model	2024	Custom protein structure	121,800 protein−small molecule structures, 112,546 protein-metal complexes, 12689 structures with covalently modified amino acids filtrate from PDB	Digoxigenin-, heme-, and bilin-binding proteins	https://github.com/baker-laboratory/rf_diffusion_all_atom	[53]

ProteinGenerator	Protein sequence	Diffusion model	2024	Protein structure and sequence simultaneous generation	Protein sequences acquired from PDB	Melittin, furin protease	https://github.com/RosettaCommons/protein_generator	[54]
SCUBA-D	Protein structure	Diffusion model	2024	Protein backbones generation	Protein structures downloaded from the PDB	Heme-binding and ras-binding proteins	https://github.com/liuyf020419/SCUBA-D.git/	[55]
P450Diffusion	Protein sequence	Diffusion model	2024	Generation of artificial P450 enzyme	226,509 natural P450 sequences	Cytochrome P450 enzyme	https://github.com/JiangLab2020/P450Diffusion	[62]
ProGen	Protein sequence	PLM	2023	Protein sequences generation	281 million non-redundant protein sequences from UniParc, pfam, UniprotKB, NCBI	Lysozyme, chorismate mutase, malate dehydrogenase	https://github.com/salesforce/progen	[63]
ZymCTRL	Protein sequence	PLM	2024	Generation of active enzymes	37M enzyme sequences annotated with EC numbers	Carbonic anhydrases, lactate dehydrogenases	https://huggingface.co/AI4PD/ZymCTRL	[64]
EVOLVEpro	Protein sequence	PLM and regression models	2024	Improving protein activity	12 deep mutational scanning datasets	Cas12f nucleases, T7 RNA polymerase	https://github.com/mat10d/EvolvePro	[67]

dehydrogenase (Fig. 3e), where 24 % of the generated sequences were soluble and retained wild-type catalytic activity *in vitro*, even with over 100 mutations. Besides, HelixGAN [42] is capable of generating new left-handed and right-handed α-helix structures at the atomic level, and by optimizing the exact conformation of hotspot residues, it generates novel α-helix structures in the latent space.

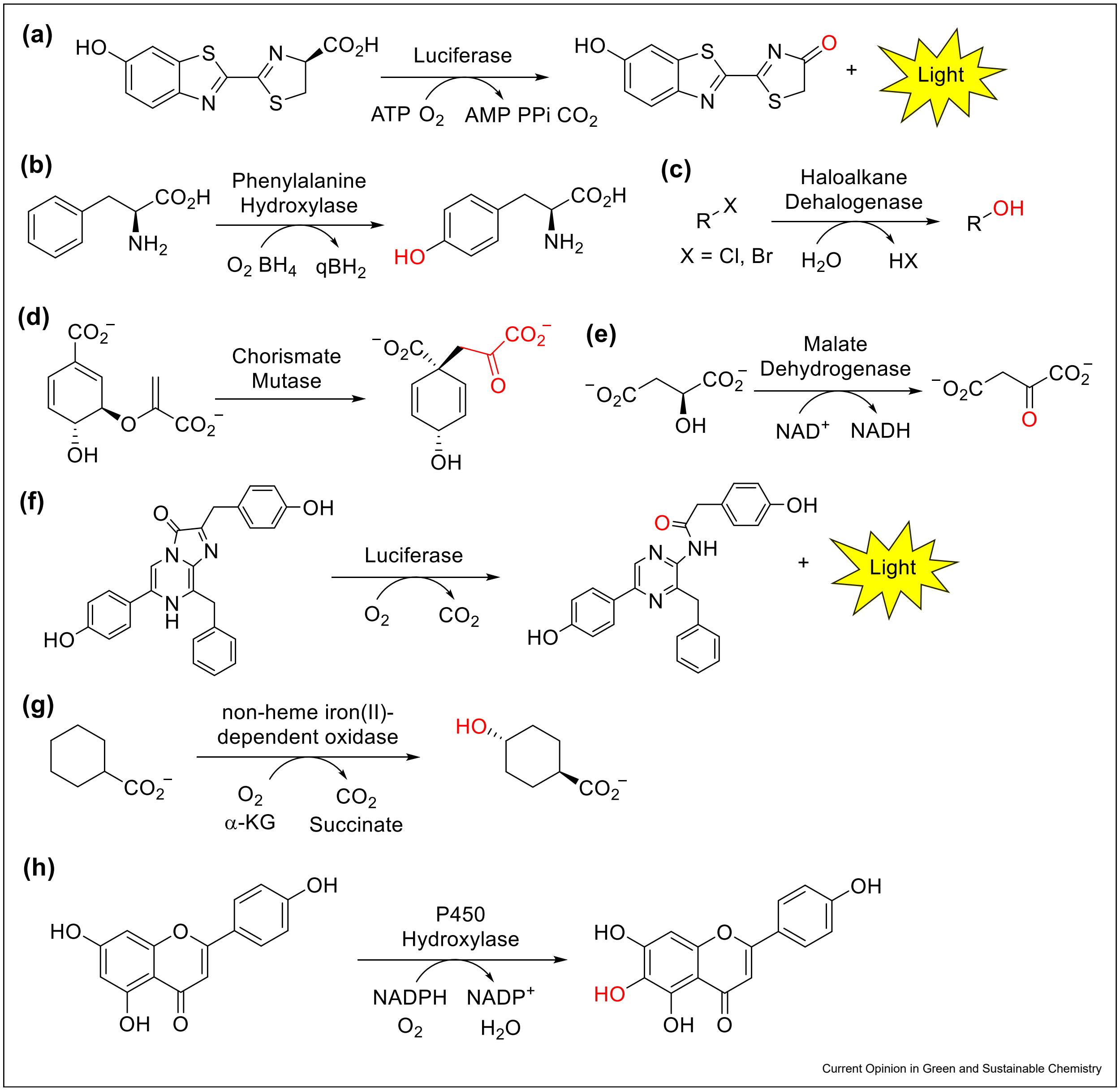
Diffusion models

Diffusion models are a family of probabilistic generative models inspired by non-equilibrium statistical physics. They progressively degrade data by introducing noise and then learn to reverse this process, enabling the generation of realistic and clean data by restoring the original, uncorrupted state. Current research on diffusion models primarily revolves around three frameworks, each offering a distinct formulation (Fig. 2c) of the forward diffusion and the backward denoising process: denoising diffusion probabilistic models (DDPMs) [43], noise conditioned score networks (NCSNs) [44], and stochastic differential equations (Score SDEs) [45].

Deep generative models such as VAEs and GANs, as mentioned above, are limited to generating small proteins or specific domains of larger proteins. Alternatively, diffusion models excel in protein design and generation, benefiting from their natural ability to effectively destroy and rebuild large, diverse protein structures. For example, Trippe et al. [46] utilized an SE (3)-equivariant GNN architecture for the motif-scaffolding generation problem, dividing the process into two stages: realistic protein backbone structure generation (ProtDiff) and motif-scaffolding via conditional sampling (SMCDiff). Unlike SMCDiff, which is a conditional generation method, FADiff [47] enables motifs to float rigidly and independently during the diffusion process, ensuring their presence while also automating motif positioning without the need for prior domain-specific expertise.

Chroma [48] and RFdiffusion [49] are also trained to tackle the motif-scaffolding challenge in protein design. RFdiffusion, a RoseTTAFold (RF) [50] based DDPM diffusion model, represents residues using Cα coordinate and N-Cα-C rigid orientation of each residue. In benchmark tests for solving the motif-scaffolding design problem, the performance of RFdiffusion is much better than RF_{joint} inpainting [51] and constrained Hallucination [52]. Moreover, by fine-tuning RoseTTAFold All-Atom for diffusion denoising tasks, RFdiffusionAA can generate folded protein structures surrounding small molecules [53]. Unlike the structure-based diffusion model RFdiffusion, ProteinGenerator [54] a sequence-space diffusion model based on RoseTTAFold, can simultaneously generate both protein sequences and structures. Additionally, SCUBA-diffusion (SCUBA-D) [55], a protein backbone denoising DDPM model trained by considering co-diffusion of sequence

Figure 3



Selected reactions catalyzed by enzymes designed via generative AI. (a) Conversion of natural D-luciferin by a luciferase [37]. (b) Hydroxylation of L-phenylalanine by a phenylalanine hydroxylase [36]. (c) Hydrolysis of haloalkanes by a haloalkane dehalogenase [38]. (d) Isomerization of chorismate by a chorismate mutase [59]. (e) Oxidation of L-malate by a malate dehydrogenase [63]. (f) Conversion of the synthetic luciferin diphenylterazine by a luciferase [71]. (g) Hydroxylation of cyclohexanecarboxylic acid by a non-heme iron(II) α-ketoglutarate-dependent oxygenase [74]. (h) Hydroxylation of apigenin by a P450 enzyme [62].

representation, competes with RFdiffusion but generates a more diverse spatial distribution of backbones, avoiding the potential biases of RFdiffusion.

Chroma, built on the framework of score SDE diffusion and graph neural network, can generate extremely large proteins and complexes (over 3000 residues), surpassing the size limit of proteins generated by several other models, including the score SDE diffusion models ProteinSGM [56] and FrameDiff [57], the DDPM diffusion models Foldingdiff [58] and Geneie [59], and the language diffusion model DiffSDS [60] and Force-Gen [61]) (all limited to <2000 residues). Different from the diffusion models mentioned before, P450Diffusion [62] focuses exclusively on the generation of P450 enzymes (P450s) by integrating the catalytic pocket design principle with a DDPM model. It is pretrained on over 226,509 natural P450 sequences and then fine-tuned using enhanced sampling, enabling the de novo generation of entirely new P450s.

Besides the above three types of DGMs, protein language models (PLMs) also perform outstandingly in protein design. ProGen [63] and ZymCTRL [64] were demonstrated successful in generating functional enzymes with low identity to natural counterparts, including lysozymes, malate dehydrogenases (Fig. 3e), chorismite mutases (Fig. 3d), carbonic anhydrases, and lactate dehydrogenases. Moreover, ProGen2 [65] suite of PLMs are scaled to 6.4B parameters, enabling the generation of novel protein sequences that adopt natural folds and are effective for zero-shot fitness prediction. PocketGen [66] employs a co-design scheme in which a bilevel graph transformer is responsible for structural encoding, and a PLM-based module handles sequence refinement, thus enabling the generation of high-fidelity protein pockets with enhanced binding affinity and structural validity. Additionally, by combining PLM with regression models, EVOLVEpro [67] can evaluate the experimental results of a small subset of mutants, iteratively update and predict the next round of mutants, thus enabling it to efficiently explore multiple active regions and avoid the issue of local optima. Its excellent performance has been demonstrated in the optimization of miniature Cas12f nucleases and a T7 RNA polymerase.

Some of these large AI models were trained on diverse datasets of various types of enzymes and proteins to ensure general applicability for different designing tasks. The success rate is much higher if the targeted enzyme group is included in the training dataset. Yet, some small AI models trained on a specific group of enzymes may perform better in specific applications (see below Section 3).

Applications of enzymes designed by generative AI

As discussed above, many generative AI models have been demonstrated to design enzymes with the same functions as their natural counterparts, including lysozymes [63], luciferases [37], recombinases [68], proteases [69], phenylalanine hydroxylases [36], haloalkane dehalogenases [38], chorismate mutases [63], malate dehydrogenases [63], carbonic anhydrases [64] and lactate dehydrogenases [64] (Fig. 3a–e), etc. From an application point of view, these designed enzymes (*e.g.*, lysozyme, luciferase, and protease) can be directly used (or act as starting templates for directed evolution) for the same applications of the natural counterparts in the fields of food, research, medical, etc. For the production or conversion of natural chemicals, the designed enzymes could be very useful if they are the bottleneck of the processes, *e.g.*, the rate-limiting enzymes in the key pathways of microbial fermentation, because they may have higher activity or are less vulnerable to allosteric regulation or inhibition by the substrates/products.

As for the production or conversion of non-natural chemicals (*e.g.*, synthesis of intermediates of synthetic pharmaceuticals, degradation of man-made plastics or pollutants), it is a more challenging task for enzyme design, because it requires high catalytic activity and/or selectivity on non-natural substrates, or even requires new types of reactions. Therefore, only the sequence information (*e.g.*, sequence-based models) is often not enough to generate efficient enzymes for these applications. Besides the traditional directed evolution, scientists have attempted to address this problem with physics-based *de novo* design [70] pioneered by David Baker. Many new enzymes were designed from the corresponding 'theozymes' in this way, yet the catalytic efficiency is often very limited without multiple rounds of evolution. Recently, a significant advance in applying generative AI for enzyme design was reported to design totally new luciferases for converting non-natural synthetic luciferins (Fig. 3f) [71]. A deep-learning-based 'family-wide hallucination' approach was employed to generate large numbers of scaffolds to accommodate the synthetic luciferin diphenylterazine, and the catalytic centers were carefully designed to stabilize the key transition state. Three designs showed luciferase activity in over 7000 experimentally tested designs. Further directed evolution was performed to tailor the pocket and thus improve the activity and specificity of the designed luciferase. In the end, the final designed luciferase has a catalytic efficiency similar to that of native luciferases, but it has a much smaller size, higher stability, and higher specificity for the synthetic luciferin. The creation of highly active luciferase for an artificial substrate using a novel scaffold demonstrated the power of generative AI in computational enzyme design. In the report, the authors also employed an improved strategy with the knowledge of the first design and ProteinMPNN [72] to create other luciferases for another synthetic luciferin 2-deoxycoelenterazine with a much higher success rate. In another report, a combination of trRosetta hallucination and ProteinMPNN was employed to design a glycoside hydrolase with an aim to significantly reduce the protein size [73]. Although several of these small designs showed soluble expression and high thermostability, they lacked detectable enzymatic activity. This highlights the complexity of enzyme catalysis and the governing chemical mechanism is still difficult to design. Very recently, ProteinMPNN was employed to generate a stabilized non-heme iron (II) α-ketoglutarate-dependent oxygenase as better starting templates for further directed evolution for non-native C–H hydroxylation of cyclohexanecarboxylic acid

(Fig. 3g) [74]. This is another significant advance in applying generative AI to design enzymes for non-native reactions. The key to the success is that the authors conducted multiple rounds of design-build-test-learn cycles, and applied more constraint (*i.e.*, fixed more residues) in the design strategies. The low success rate of functional enzyme design was also observed in another recent study when evaluating three different generative models in designing malate dehydrogenase and copper superoxide dismutase [75]. The authors then tested multiple computational metrics to develop a computational filter to improve the rate of success. Clearly, such a filter could be a general tool to benchmark different generative models and improve the success rate of computational enzyme design. Another strategy for effective design is developing specific models for a certain type of enzyme instead of a big model for all the proteins, as demonstrated in a recently developed P450Diffusion model [62]. The authors created several novel artificial P450 enzymes for converting apigenin into scutellarein with a higher catalytic capacity than the natural counterpart (Fig. 3h). Although the reaction tested is a natural conversion, the designed P450 enzymes could be also useful for other substrates or type of reactions due to the high catalytic promiscuity of this type of enzyme.

Conclusions and outlook

A very important trend in the research field of chemistry and biochemistry is integrating cutting-edge information technology into modern biotechnology, with computational protein design and protein structure prediction awarded with the 2024 Nobel Prize in Chemistry. Nowadays, AI-based machine learning is dramatically changing how we discover and engineer enzymes for the production and conversion of chemicals and other applications. More excitingly, AI-based methods could create novel and functional enzymes. Very recently, many generative models, including VAEs, GANs, and diffusion models, have been reported for protein design purpose, and some of them were experimentally validated for certain types of proteins and enzymes. Many of these AI tools are freely available to users to design new enzymes, yet a thorough and fair benchmarking of different methods is still lacking. In particular, the rate of functional design is often highly compromised especially when they are employed by researchers other than the developers. Certain constrains, such as a computational filter with multiple metrics or insights from multiple rounds of the design-build-test-learn cycle, could significantly improve the scenarios of computational design. Another strategy is to develop customized models for a certain group of enzymes, which will be of high interest to certain researchers and industries. Nevertheless, most recently AI-designed enzymes are still limited to catalyzing the existing reactions of their natural counterparts. For academic challenges and chemical industrial applications, accurately designing efficient and selective enzymes with new catalytic functions (*e.g.*, non-natural substrates, or even new types of reactions) is highly desired but remains to be achieved. We believe that recently developed All-Atom models [23,53] taking account of the structures of both proteins and related molecules, will significantly contribute to solving this grand challenge in protein design.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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